

Lab 00 — Glossary

Plain-language definitions of the tools and terms you'll meet in the labs

How to Use This Glossary

Keep this page open while you work. If you come across a term you don't know, look it up here, or ask Claude to explain it: that's part of learning as you build.

Terms

Term	What it means	Where you'll see it
AI coding tool	An AI assistant that writes and changes code for you from plain-language instructions. In this workshop, that's Claude in Cowork.	All labs
AI model	The trained AI "brain" that reads text (or images) and produces answers. Some run in the cloud (like Claude); others run on your laptop (through Ollama).	All labs
API (application programming interface)	A way for one program to talk to another. Your Lab 02 app uses the Webex API to read your spaces and messages.	Lab 02
Bot token	An access token for a Webex bot account. Bots can't read a user's full conversation history, so Lab 02 uses a personal access token instead.	Lab 02
ChromaDB	A local database that stores embeddings and finds the chunks whose meaning is closest to your question.	Lab 01
Chunk / chunking	Splitting a document into small, overlapping pieces of text so the most relevant pieces can be found and sent to the model.	Lab 01
Citation	A note showing where an answer came from, such as the file name and page number.	Lab 01

Term	What it means	Where you'll see it
Claude	Anthropic's AI assistant. You use it through the Claude desktop app to build the labs.	All labs
Claude desktop app	The Claude application installed on your laptop. You log in with your organization account and start Cowork sessions from it.	Lab 00
Cloud model	An AI model that runs on a company's servers over the internet. The lab apps deliberately avoid sending your data to cloud models.	All labs
Command	An instruction you type into Terminal and run by pressing Return, such as <code>ollama --version</code> .	All labs
Confidence score	How sure the model says it is about a result, from 0% to 100%. It's a hint, not proof: always check the evidence.	Lab 02
Confirmation	In Lab 03, the review checkbox plus the Confirm event button. Nothing can be downloaded until you confirm, and any edit clears the confirmation.	Lab 03
Context rich	After finishing the labs, your Claude session "remembers" what you've built and how you work, which makes it a strong starting point for your own projects.	All labs
Context window	How much text (and images) a model can read and write in one request, measured in tokens.	Lab 03
Cowork	The mode of the Claude desktop app you use for the labs. Claude can work with files in a folder you connect.	Lab 00
Daylight saving time (DST)	Clocks moving forward or back an hour. Some times don't exist or happen twice on those days, so Lab 03 asks you to resolve them instead of guessing.	Lab 03
Dependencies	The add-on libraries ("packages") an app needs to run. uv installs them for you.	All labs

Term	What it means	Where you'll see it
Embedding	A list of numbers that represents the meaning of a piece of text. Texts about similar things get similar numbers, which lets the app search by meaning rather than exact words.	Lab 01
Embedding model	A model that turns text into embeddings. Lab 01 uses <code>gpt4o-mini-embed-text</code> .	Lab 01
.env file	A small settings file in a project folder, holding values such as your Webex access token. Its name starts with a dot, so Finder hides it.	Lab 02
Evidence	The original text (a PDF passage or a Webex message) that an AI result is based on, shown so you can check it yourself.	Labs 01, 02
Extraction	Pulling structured details (like tasks, dates or times) out of text or images.	Labs 02, 03
gemma3:12b	A Google AI model you can run in Ollama. An alternative model option in Lab 02.	Lab 02
HEIC / HEIF	The photo format many iPhones use by default. Supporting it is an optional improvement in Lab 03.	Lab 03
IANA time zone	The standard name for a time zone, such as <code>America/New_York</code> or <code>Europe/London</code> . Lab 03 requires one for every event.	Lab 03
.ics file	A standard calendar-event file that Outlook, Google Calendar and Apple Calendar can import.	Lab 03
Inference	The act of an AI model producing an answer. "Local inference" means the model runs on your laptop.	All labs
JSON	A simple, structured text format that programs can read easily. The lab apps ask the model to reply in JSON.	Labs 02, 03
llama3.2:3b	A small, fast AI model from Meta that writes the answers in Lab 01.	Lab 01

Term	What it means	Where you'll see it
LLM (large language model)	An AI model trained on large amounts of text that can understand and write language. Claude, Llama and Qwen are LLMs.	All labs
localhost	Your own computer, as seen by your web browser. <code>http://localhost:8501</code> means "the app running on this laptop, at port 8501".	All labs
Markdown (.md)	A simple text format for documents, where symbols such as <code>#</code> and <code>**</code> add headings and bold. The lab guides and READMEs are Markdown files.	All labs
Menu bar	The strip across the top of a Mac screen. The llama icon there means Ollama is running.	Lab 00
Mocked test	A test that uses a pretend version of something (like the AI model), so it runs quickly without needing the real thing.	All labs
MVP (minimum viable product)	A simple, working first version that does the main job. Each lab builds an MVP you can then improve.	All labs
nomic-embed-text	A small embedding model you run in Ollama to turn document text into embeddings.	Lab 01
OCR (optical character recognition)	Technology that reads text from a picture, such as a scanned page. The Lab 01 app doesn't do OCR, so scanned PDFs can't be read.	Lab 01
Ollama	A free, open-source app that downloads and runs AI models on your own computer.	All labs
Opus 5.5	The Claude model you use in Cowork for the labs.	Lab 00
Personal access token	A secret code from developer.webex.com that lets your app read your Webex spaces as you. It expires after 12 hours; keep it private.	Lab 02
Pillow	A Python library for opening, checking and resizing images.	Lab 03

Term	What it means	Where you'll see it
Port	A numbered "door" an app uses on your computer. Each lab app has its own (8501, 8502, 8503), so all three can run at the same time.	All labs
Prompt	The instructions you give an AI in plain language. Each lab starts with a prompt that describes the app to build.	All labs
Prompt injection	Text that tries to trick an AI into following it as instructions, for example words hidden in an image. The Lab 03 app treats image text as data only.	Lab 03
Pydantic	A Python library that checks data has the expected shape. Lab 03 uses it to define and validate the model's answers.	Lab 03
PyMuPDF	A Python library that reads the text from PDF files, page by page.	Lab 01
pytest	A tool that runs a project's automated tests. You run it with <code>uv run pytest</code> .	All labs
Python	The programming language all three lab apps are written in.	All labs
qwen3.5 (4b, 9b, 27b)	A family of AI models from Alibaba that can read text and images. <code>9b</code> is the default for Labs 02 and 03; <code>4b</code> is smaller and faster; <code>27b</code> is larger and slower.	Labs 02, 03
RAG (retrieval-augmented generation)	A technique where the app first finds (retrieves) the most relevant passages from your documents, then has the model write (generate) an answer from them.	Lab 01
Read-only	Allowed to read but never change anything. The Lab 02 app never sends, edits or deletes anything in Webex.	Lab 02
README	A file (usually <code>README.md</code>) that explains what a project does and how to install, start, use and troubleshoot it. Claude writes one for each lab app.	All labs

Term	What it means	Where you'll see it
Schema / structured output	A strict description of the shape an answer must have (fields and types). Ollama can force a model's reply to match it, so the app can read it reliably.	Labs 02, 03
Session memory	Data kept only while the browser tab is open. In Lab 03, drafts can be lost if you refresh the page or restart the app.	Lab 03
SQLite	A small database stored in a single file on your laptop. The lab apps use it to remember documents, messages, tasks and your decisions.	Labs 01, 02
Streamlit	A Python library for building simple web apps. All three lab apps use it for their interface.	All labs
Terminal	The Mac app where you type commands. Open it with ⌘ + Space , type Terminal , and press Return .	All labs
Token (AI)	A small piece of text (roughly part of a word) that AI models read and write. Context windows are measured in tokens.	Lab 03
Transcription	The text a model reads from an image, written out. It can contain mistakes, so check it against the image.	Lab 03
uv	A fast, free tool that installs Python and each app's dependencies, keeping every project's setup separate.	All labs
uv run	Runs a command inside the project's own environment, e.g. <code>uv run streamlit run app.py</code> starts the app.	All labs
uv sync	Installs everything a project needs, as listed in its <code>pyproject.toml</code> file.	All labs
Vector database	A database that stores embeddings and quickly finds the most similar ones. ChromaDB is a vector database.	Lab 01
Vibe coding	Building software by describing what you want to an AI and refining the result together, rather than writing every line of code yourself.	All labs

Term	What it means	Where you'll see it
Vision model	An AI model that can read images as well as text. Lab 03 uses one to read event flyers.	Lab 03
Webex space	A Webex chat room, either a one-to-one conversation or a group.	Lab 02